

Suresh Gundumogula

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GitHub | LinkedIn | Portfolio

SUMMARY

Information Technology graduate specializing in AI engineering, Retrieval-Augmented Generation (RAG), and intelligent full-stack systems. Experienced in building end-to-end AI applications involving semantic retrieval, vector search, reranking pipelines, and machine learning models using Python, FastAPI, React, and modern LLM tooling. Strong foundation in ML systems, backend engineering, and applied AI architecture.

SKILLS

Programming Languages: Python, JavaScript, SQL

AI & ML Frameworks: PyTorch, Scikit-learn, LangChain, Sentence Transformers, Pandas, NumPy

AI & ML Concepts: Retrieval-Augmented Generation (RAG), Convolutional Neural Networks (CNNs), Semantic Search, Vector Embeddings, Cross-Encoder Reranking, Feature Engineering

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs

Frontend: React.js, Tailwind CSS, HTML, CSS

Databases & Vector Stores: MongoDB, MySQL, ChromaDB

DevOps & Tools: Git, GitHub, Postman, Vercel, Netlify, Render, OpenRouter API

EXPERIENCE

MERN Stack Intern | UptoSkills (Virtual)

May 2025 – July 2025

- Engineered responsive React components and integrated them with backend REST APIs for HRMS Social Connect, a MERN-based networking platform.
- Developed and optimized backend APIs utilizing Node.js, Express.js, and MongoDB to enhance platform functionality and secure data flow.

PROJECTS

RepoGPT — Code Intelligence RAG System | *Python, FastAPI, LangChain, ChromaDB, React*

GitHub

- Built a repository-level Code RAG system that clones GitHub repositories, indexes codebases, and answers natural language questions with cited source code retrieval.
- Designed a two-stage retrieval pipeline combining semantic vector search (top-10 recall) with Cross-Encoder reranking (top-3 precision), achieving 75% retrieval accuracy on benchmark evaluation queries.
- Implemented language-aware code chunking for 12 programming languages using LangChain splitters, ChromaDB vector storage, and Sentence Transformer embeddings.
- Developed a full-stack AI application with FastAPI REST backend and React chat interface featuring markdown rendering, syntax-highlighted code blocks, and repository-aware code retrieval.

Mirror Chess — Personalized Chess AI Bot | *PyTorch, Python, FastAPI, React*

GitHub | Live

- Trained a CNN-based policy network on 8,000+ personal games to model and replicate player-specific decision patterns.
- Designed a hybrid inference system combining neural network predictions with Minimax (alpha-beta pruning) to enforce tactical correctness and prevent blunders.
- Built and deployed a full-stack application with a FastAPI backend for low-latency inference and a responsive React frontend for real-time gameplay.

F1 Strategy Simulator | *Python, Scikit-learn, Pandas, FastAPI, React*

GitHub | Live

- Built an end-to-end race simulation system to predict lap times and evaluate pit stop strategies across full races.
- Developed event-aware ML models to handle different race phases (normal laps, pit stops, Safety Car conditions).
- Engineered temporal features such as tyre degradation, fuel load, and stint progression to capture race dynamics.
- Achieved ~0.5s MAE on lap time prediction and deployed an interactive dashboard.

EDUCATION

B.Tech in Information Technology | *JNTU-GV College of Engineering, Vizianagaram*

2023 – 2026

Diploma in Computer Engineering | *BVC Engineering College, Amalapuram*

2020 – 2023

Secondary School Education (SSC) | *ZPHS, I. Polavaram*

2019 – 2020